

	A	B	C	D	E	F	G	H	I	J
1	From the data given below, test whether the average expenditure per day of a family is 1500 per day									
2	at 90% confidence level.									
3	1250	1400	1850	2000	2200	1750	1950	1900	1200	1000
4										
5	Solution:- Here, We have									
6		Sample size (n) =		10	=COUNT(A3:J3)					
7		Pop. Mean (μ) =		1500						
8		Sample Mean ($\bar{x}$) =		1650	=AVERAGE(A3:J3)					
9		Sample S.D.(S) =		404.832	=STDEV.S(A3:J3)					
10										
11	Here, We set up hypothesis as									
12	$H_0 : \mu = 1500$ i.e. Average expenditure of a family is Rs. 1500.									
13	$H_1 : \mu \neq 1500$ i.e. Average expenditure of a family is other than Rs. 1500.									
14										
15		S.E($\bar{x}$) =		128.019	=D9/SQRT(D6)					
16		$t_{cal.}$ =		1.1717	=(D8-D7)/D15					
17		C.I. (1- α) =		90%						
18	Level of significance (α) =		0.1	=1-D17						
19		d.f. =		9	=D6-1					
20		$t_{tab.}$ =		1.83311	=T.INV.2T(D18,D19)					
21										
22	Decision:- Since, $t_{cal.} < t_{tab.}$, we accept H_0 and reject H_1 with the conclusion									
23	that average expenditure of a family is Rs. 1500.									
24										
25										
26										
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28										
29										
30	Name = Hari Bashyal									